



NATIONAL TECHNICAL UNIVERSITY OF ATHENS
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
DIVISION OF COMPUTER SCIENCE

Continuous Machine Learning for Cooperative, Connected and Automated Mobility applications

DIPLOMA THESIS

of

Georgios A. Kyriakopoulos

Supervisor: Panayiotis Tsanakas
Professor

Athens, October 2025



NATIONAL TECHNICAL UNIVERSITY OF ATHENS
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
DIVISION OF COMPUTER SCIENCE

Continuous Machine Learning for Cooperative, Connected and Automated Mobility applications

DIPLOMA THESIS

of

Georgios A. Kyriakopoulos

Supervisor: Panayiotis Tsanakas
Professor

Approved by the three-member committee of inquiry in October 2025

.....
Panayiotis Tsanakas
Professor NTUA

.....
Andreas-Georgios Stafylopatis
Professor Emeritus NTUA

.....
Georgios Alexandridis
Assistant Professor NKUA

Athens, October 2025

Copyright

Περίληψη

Γράψτε εδώ την ελληνική περίληψη της διπλωματικής σας εργασίας.

Περιγράψτε συνοπτικά:

- Το πρόβλημα που αντιμετωπίζετε
- Τη λύση που προτείνετε
- Τα κύρια αποτελέσματα
- Τη συνεισφορά της εργασίας

Λέξεις Κλειδιά: Λέξη κλειδί 1, Λέξη κλειδί 2, Λέξη κλειδί 3, Λέξη κλειδί 4

Abstract

Write your English abstract here.

Briefly describe:

- The problem you are addressing
- Your proposed solution
- Main results and contributions
- Impact and significance

Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4

Acknowledgements

I would like to thank...

[Write your acknowledgements here. Usually thank your supervisor, committee, collaborators, friends, and family.]

Contents

Contents	10
List of Figures	11
List of Tables	13
Extended Abstract in Greek	15
1 Introduction	17
1.1 Motivation	17
1.2 Problem Statement	17
1.3 Contributions	17
1.4 Thesis Organization	17
2 Background and Related Work	19
2.1 Background	19
2.2 Related Work	19
2.3 Summary	19
3 Methodology	21
3.1 Overview	21
3.2 Dataset Generation	21
3.3 Model Architecture	21
3.4 Drift Detection	21
4 Implementation	23
4.1 Platform Architecture	23
4.2 Technology Stack	23
4.3 Implementation Details	23
5 Evaluation and Results	25
5.1 Experimental Setup	25
5.2 Results	25
5.3 Discussion	25
6 Conclusion and Future Work	27
6.1 Summary	27
6.2 Limitations	27

6.3 Future Work	27
Bibliography	29

List of Figures

List of Tables

Extended Abstract in Greek

Εισαγωγή

Γράψτε μια σύντομη εισαγωγή στο πρόβλημα...

Θεωρητικό Υπόβαθρο

Περιγράψτε το θεωρητικό υπόβαθρο...

Μεθοδολογία

Περιγράψτε την προσέγγισή σας...

Υλοποίηση

Περιγράψτε την υλοποίηση...

Αποτελέσματα

Παρουσιάστε τα αποτελέσματα...

Συμπεράσματα

Συνοψίστε τη συνεισφορά σας...

Chapter 1

Introduction

1.1 Motivation

Why is this problem important?

1.2 Problem Statement

What problem are you solving?

1.3 Contributions

What did you contribute?

- Contribution 1
- Contribution 2
- Contribution 3

1.4 Thesis Organization

Brief overview of chapters.

[1]

Chapter 2

Background and Related Work

2.1 Background

Provide necessary background.

2.2 Related Work

Review related work.

2.3 Summary

Summarize and position your work.

Chapter 3

Methodology

3.1 Overview

High-level overview of your approach.

3.2 Dataset Generation

How you generated data with SUMO, etc.

3.3 Model Architecture

Your ML model design.

3.4 Drift Detection

Concept drift detection approach.

Chapter 4

Implementation

4.1 Platform Architecture

System design and components.

4.2 Technology Stack

Tools, frameworks, languages used.

4.3 Implementation Details

Key implementation decisions and challenges.

Chapter 5

Evaluation and Results

5.1 Experimental Setup

Datasets, metrics, baselines.

5.2 Results

Present your experimental results.

5.3 Discussion

Analyze and interpret the results.

Chapter 6

Conclusion and Future Work

6.1 Summary

Recap your contributions.

6.2 Limitations

Honest assessment of limitations.

6.3 Future Work

What could be done next?

Bibliography

- [1] Doe, J. and Smith, J. “An Example Paper”. In: *Journal of Examples* 1.1 (2024).